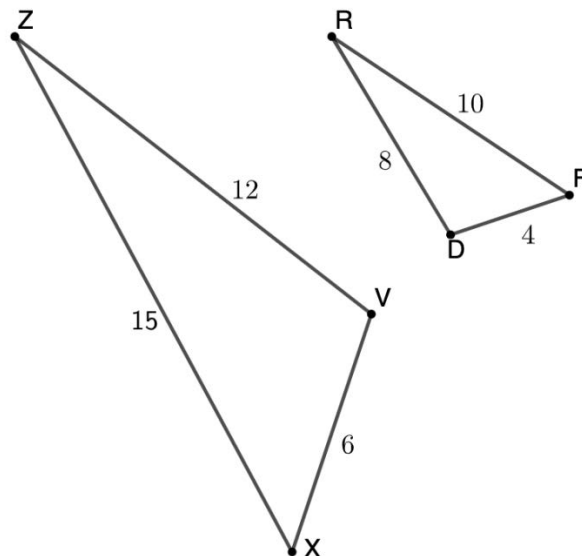


Geometry
Unit 6
Lesson 8 Homework

Name Answer Key
Date _____
Period _____

$\triangle ZVX$ and $\triangle RFD$ are shown.



1. Complete the table.

Measurements of $\triangle ZVX$	Measurements of $\triangle RFD$	Ratio
$ZX = 15$	$RF = 10$	$\frac{ZX}{RF} = \frac{15}{10} = \frac{3}{2}$
$XV = 6$	$FD = 4$	$\frac{XV}{FD} = \frac{6}{4} = \frac{3}{2}$
$VZ = 12$	$DR = 8$	$\frac{VZ}{DR} = \frac{12}{8} = \frac{3}{2}$

2. What similarity theorem can be applied to show that the triangles are similar?

Side-Side-Side Similarity

3. Write the similarity statement.

$\triangle ZVX \sim \triangle RFD$

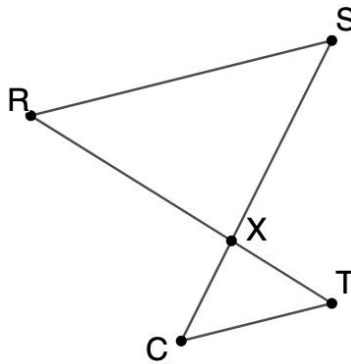
4. Complete each.

$$\frac{ZV}{XV} = \frac{RD}{FD}$$

$$\frac{RF}{ZX} = \frac{DF}{VX}$$

$$\frac{XZ}{VZ} = \frac{FR}{DR}$$

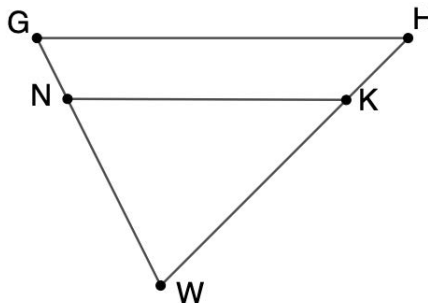
$\triangle RSX$ and $\triangle XTC$ are shown where $\angle RSX \cong \angle TCX$.



Complete the two-column proof.

	Statement	Reason
	$\angle RSX \cong \angle TCX$	Given
5.	$\angle SXR \cong \angle CXT$	Vertical angles are congruent
6.	$\triangle RSX \sim \triangle \underline{\hspace{2cm}}$	AA ~

$\triangle GWH$ and $\triangle NWK$ are shown.



Given: $WN = \frac{2}{3}WG$ and $WK = \frac{2}{3}WH$

Prove: $\triangle GWH \sim \triangle NWK$

	Statement	Reason
	$WN = \frac{2}{3}WG$ and $WK = \frac{2}{3}WH$	Given
7.	$\underline{\hspace{1cm}} = \frac{2}{3}$ and $\underline{\hspace{1cm}} = \frac{2}{3}$	Division Property of Equality
8.	$\frac{WN}{WG} = \frac{WK}{WH}$	Transitive Property of Equality
9.	$\angle GWH \cong \angle NWK$	Reflexive Property of Congruence
10.	$\triangle GWH \sim \triangle NWK$	SAS ~